MIL-W-23554C(SH)
16 March 1984
SUPERSEDING
MIL-W-23554B(SHIPS)
17 August 1967
(See 6.5)

MILITARY SPECIFICATION

WASHER-EXTRACTOR, LAUNDRY, COMMERCIAL (SHIPBOARD)

This specification is approved for use by the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a 16-pound (minimum) dry weight capacity automatic combination washer-extractor machine for use aboard ships.

1.2 Classification. The washer-extractors shall be of the following types, as specified (see 6.1):

Type I - For submarines.
Type II - For surface ships.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified, the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

QQ-N-281 - Nickel-Copper Alloy Bar, Rod, Plate, Sheet, Strip, Wire, Forgings, and Structural and Special Shaped Sections.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Sea Systems Command, SEA 55Z3, Department of the Navy, Washington, DC 20362 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 3510

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MILITARY

MIL-S-901 - Shock Tests, H.I. (High-Impact); Shipboard Machinery, Equipment and Systems, Requirements For.

MIL-L-3153 - Laundry and Dry Cleaning Machinery and Equipment (For Fixed Installations), Preparation For Delivery Of.

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables For Inspection By Attributes.

MIL-STD-740 - Airborne and Structureborne Noise Measurements and Acceptance Criteria Of Shipboard Equipment.

DOD-STD-1399, Section 300 - Interface Standard For Shipboard Systems. Electric Power, Alternating Current (Metric).

(Copies of specifications and standards required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

UNIFORM CLASSIFICATION COMMITTEE AGENT

Uniform Freight Classification Ratings, Rules and Regulations

(Application for copies should be addressed to the Uniform Classification Committee Agent, Tariff Publication Officer, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

THE INSTITUTE OF ELECTRICAL AND ELECTRONICS ENGINEERS, INC. (IEEE)

IEEE STD 45 - IEEE Recommended Practice for Electric Installations on Shipboard. (DoD adopted)

(Application for copies should be addressed to the Institute of Electrical and Electronics Engineers, 345 East 47th Street, New York, NY 10017.)

AMERICAN IRON AND STEEL INSTITUTE (AISI)

Steel Products Manual.

(Application for copies should be addressed to the American Iron and Steel Institute, 350 Fifth Ave., New York, NY 10017.)

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ASTM

A 269 - Standard Specification for Seamless and Welded Austenitic Stainless Steel Tubing for General Service. (DoD adopted)

B 43 - Standard Specification for Seamless Red Brass Pipe, Standard Sizes. (DoD adopted)

B 61 - Standard Specification for Steam or Valve Bronze Castings. (DoD adopted)

B 62 - Standard Specification for Composition Bronze or Ounce Metal Castings. (DoD adopted)

B 75 - Standard Specification for Seamless Copper Tube. (DoD adopted)

D 2000 - Standard Classification System for Rubber Products In Automotive Applications. (DoD adopted)

(Application for copies should be addressed to the ASTM, 1916 Race Street, Philadelphia, PA 19103.)

(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 First article. When specified (see 6.1.1), a sample shall be subjected to first article inspection (see 4.3 and 6.2).

3.2 Materials.

3.2.1 Structure. Frame, base, and strength members shall be constructed of malleable iron, ductile iron or steel, hot dipped galvanized for corrosion protection. Gray cast iron or other brittle materials shall not be used for these parts.

3.2.2 Shell. Shell shall be made of corrosion-resisting steel in accordance with AISI type 302 or 304 or nickel-copper alloy in accordance with QQ-N-281. Heads of the shell shall be of corrosion-resisting metal or ferrous metal with corrosion-resisting metal liners on the side exposed to the washing solution. The shell shall be provided with an aluminum alloy or corrosion-resisting steel door frame.

3.2.3 Cylinder. The cylinder shall be made of corrosion-resisting steel, in accordance with AISI type 302 or 304, or of monel (nickel-copper alloy, minimum 60 percent nickel). The driving trunnion and spider may be of the same material as the cylinder, or of ferrous metal protected by hot dip galvanizing or a corrosion-resisting coating.

3.2.4 Cabinet. The cabinet shall be made of corrosion-resisting steel in accordance with AISI type 302 or 304.

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3.2.5 Metal joints. Rivets, bolts, screws, nuts and washers shall be of steel, except when brass or corrosion-resisting metal is fastened, in which case they shall be of brass or corrosion-resisting metal. Where dissimilar metals are fastened, rivets, bolts, screws, nuts and washers shall be galvanically compatible with materials being joined. Electrodes or welding rods used for welding shall deposit material similar to the base metal.

3.2.6 Pipes, fittings and valves. Parts in contact with the wash solution shall be bronze or corrosion-resisting steel. The following materials shall be used:

Pipes:	Corrosion-resisting steel, in accordance with ASTM A 269, grade TP 302 or 304.
	Brass, in accordance with ASTM B 43, grade A.
	Copper, in accordance with ASTM B 75.
Fittings:	Bronze, in accordance with ASTM B 61 or B 62.
Valves:	Cast bronze, in accordance with ASTM B 61.

3.2.7 Recovered materials. Unless otherwise specified herein, all equipment, material, and articles incorporated in the products covered by this specification shall be new and shall be fabricated using materials produced from recovered materials to the maximum extent practicable without jeopardizing the intended use. The term "recovered materials" means materials which have been collected or recovered from solid waste and reprocessed to become a source of raw materials, as opposed to virgin raw materials. None of the above shall be interpreted to mean that the use of used or rebuilt products is allowed under this specification unless otherwise specifically specified.

3.3 Design. The type I and type II machines shall be front loading, automatic, shell and cylinder (open jacket) type with a rotating and reversing action about its horizontal axis. The machines shall be 16 pounds minimum dry weight capacity. Each machine shall be a complete unit, enclosed within a cabinet and consist of necessary controls, valves, motor and appurtenances for automatically washing, rinsing and extracting clothing as specified herein.

3.3.1 Special design requirements - type I washer-extractor only. The type I machine shall be so constructed that it can be readily disassembled into sections or parts that can be passed through a 25-inch diameter submarine hatch and reassembled with a minimum of effort in the submarine laundry space. Cutting of parts by torch or similar means to accomplish disassembly at the site and reassembly of parts by welding or similar method, shall not be used to meet disassembly requirement. The automatic programmer, (see 3.3.2.11) shall be constructed for remote mounting by installing activity and shall not be installed on washer-extractor. The programmer shall be provided with a minimum 20 foot electrical cable for connecting the programmer to the washer-extractor. The terminal ends of connecting wiring shall be provided with connectors and receptacles for convenient installation. Connectors shall be lock type to ensure secure connection.

3.3.2 Detail requirements. Except as noted in the following paragraphs, all the following detail requirements apply to both type I and type II washer-extractors.

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3.3.2.1 Size. The type I machine overall dimensions shall not be more than 34 inches in height, 24 inches in width (across front) and 28 inches in depth (front to back). The type II machine overall dimensions shall not be more than 51 inches in height, 28 inches in width (across front) and 28 inches in depth (front to back).

3.3.2.2 Weight. The machine, including controls, programmer and appurtenances, shall not weigh more than 400 pounds.

3.3.2.3 Water consumption. Water consumption shall not exceed 50 gallons for a maximum rated capacity load.

3.3.2.4 Mounting. The base shall be fitted with bolt holes (not less than four) for anchoring the unit to the deck. Size of bolt holes shall be suitable for 5/8-inch bolt.

3.3.2.5 Shell. The shell shall be watertight, except in way of required openings, and rigid to prevent distortion or objectionable vibration under normal shipboard operation. Reinforcement shall be provided where necessary to properly support attachments. Provision shall be made for replacement of cylinder. Shell seams shall not pass through the drain valve opening.

3.3.2.5.1 Shell openings. The shell shall have openings for a door, drain valve, water inlet valves, cylinder shaft and overflow connection. No other openings shall be provided.

3.3.2.6 Cylinder. The cylinder shall be rigid to prevent distortion or failure from stresses caused during the extraction cycle at rated capacity. Perforations in the cylinder shall be sized and spaced for efficient washing, rinsing and extraction. Perforations shall be free from burrs or imperfections that will adversely affect clothing.

3.3.2.7 Door. The shell shall be fitted with a round door mounted on heavy duty hinges, complete with neoprene gasket and heavy duty latch to ensure watertightness when door is closed. Where door is fitted with a window, the glass used shall be of a quality and thickness to prevent damage under normal operation. The glass shall be sealed and capable of ready replacement.

3.3.2.8 Drain valve. The drain valve shall have a replaceable resilient seat and shall be fitted at the lowest point of the shell. The valve shall be normally open.

3.3.2.9 Piping. Piping, valves and fittings shall be capable of withstanding a working pressure of 100 pounds per square inch (lb/in^2).

3.3.2.9.1 Inlet valves. Fresh water inlet valves shall be connected to the machine, a minimum of 3 inches above the maximum high-water level.

3.3.2.9.2 Overflow connection. Machine shall be fitted with an overflow connection. The connection shall be of a size to prevent water from rising 3 inches above high-water level when both hot and cold water valves are fully opened and inlet pressure is 40 lb/in^2.

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3.3.2.10 Drive. The cylinder shall be driven by a two speed electric motor as specified in 3.3.3.1 through V-belts to provide proper speeds for washing and extracting cycles. The cylinder shall reverse rotation a minimum of three times per minute during the washing cycle.

3.3.2.10.1 Bearings. The cylinder shall be supported by antifriction bearings (see 3.3.3.1) which shall be readily replaceable and provided with seals to prevent water entry. Bearings shall be provided in accordance with bearing contractor recommendation and shall be selected based upon the maximum bearing load during operation of the washer-extractor.

3.3.2.10.2 V-belts. V-belts shall have a minimum service life of 5000 hours of machine operation. A simple means shall be provided for properly tensioning the V-belts.

3.3.2.10.3 Shafts. Shafts and related drive items shall be constructed so that the maximum stresses do not exceed 70 percent of the yield strength of the materials under the maximum applied operating torque under full load operating conditions.

3.3.2.11 Automatic programmed control. The machine shall be equipped with a timer-programmer, preprogrammed by the manufacturer, to automatically control the fill, wash, drain, rinse and extract operations of the machine from beginning to end of entire machine cycle. The control shall be programmed to provide at least one wash, three rinses, two intermediate and one final extract operation.

3.3.2.12 Water temperature control. Temperature switches shall be provided to allow operator selection of three wash water and two rinse water temperatures as follows: hot wash - warm rinse, warm wash - warm rinse, warm wash - cold rinse, cold wash - cold rinse.

3.3.2.13 Water level control. An automatic water level control shall be provided to control at least two water levels. The control shall automatically maintain high or low water levels as selected. Dials or switches shall be provided for operator selection of water level. The water level controller and washer-extractor timer shall be electrically interlocked so that timer does not advance during water filling operation.

3.3.2.14 Lubrication. Grease fittings with extension lines leading to the point of lubrication shall be conveniently located on front or side of machine to facilitate machine lubrication from front of machine. Other points of lubrication shall be easily accessible.

3.3.2.15 Finish and color. The washer-extractor shall be finished according to contractors standard commercial practice.

3.3.3 Electrical requirements. Unless otherwise specified in the contract or order, all electrical equipment shall conform to the recommended practices for electrical installation aboard ship in accordance with IEEE-45. Electrical equipment shall meet the constraints and be constructed for operation on a type I electrical system in accordance with DOD-STD-1399, section 300. The control panel shall be dripproof. Pushbuttons and switches shall be arranged for safe and convenient operation.

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3.3.3.1 Motor. Motor shall be two speed and conform to the following requirements:

Voltage:	440, 3 phase, 60 hertz
Type:	Squirrel cage induction
Ambient temperature:	40 degrees Celsius
Duty:	Wash; reversing, intermittent, 3 reversals (minimum) per minute. Extract; 1 hour full load operation followed by 10 minute rest.
Bearings:	Ball.

3.3.3.2 Controls. Motor controllers shall be of the low voltage protection type. Electrical components mounted outside the control panel shall be provided with dripproof enclosures.

3.3.3.3 Wiring methods and practices. Terminal blocks shall be plainly marked to correspond with the markings on the electrical diagram. Wires run in conduit or in harness assemblies shall be marked to correspond with the electrical diagram.

3.3.4 Safety features. Exposed belts, shafts, pulleys and other moving parts shall be fully enclosed by metal guards. Metal guards shall be firmly supported but readily removable for maintenance.

3.3.4.1 Door interlock. An electric interlock shall be provided to prevent opening door when machine is operating.

3.3.4.2 Emergency stop. A means shall be provided for readily stopping the machine in an emergency.

3.4 Performance. The washer-extractor shall meet the performance requirements in 3.4.1 through 3.4.4.

3.4.1 Inclined operation. The machine shall function without reduction in performance or spill more than a total of one gallon of water when operated at an angle of 12 degrees to each side of two vertical planes at right angles to each other when operated (see 4.3.2). Spillage, if any, shall not cause damage to electrical parts.

3.4.2 Airborne noise and structureborne vibration, type I washer-extractor only. The washer-extractor shall not exceed the acceptable airborne and structureborne vibration limits specified in tables II and III when tested in accordance with 4.3.3 through 4.3.3.7.

3.4.3 Extraction. The machines shall be capable of extracting water from saturated test material to a maximum water retention rate of 80 percent of the normal dry weight (for instance 16 pounds of test material saturated and extracted shall weigh no more than 28.8 pounds after 5 minutes extraction). Test cloth shall contain no less than 80 percent cotton.

3.4.4 Shock. No part of the washer-extractor or the automatic programmer control unit shall become adrift when tested to the high impact shock tests in accordance with MIL-S-901, grade B as specified in 4.3.5.

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3.5 Identification and information plates. An identification plate of corrosion resistant steel or brass and an information plate of corrosion resistant steel or brass shall be secured in a conspicuous place on each machine. Marking shall be of a method which shall be permanent and legible (see 3.5.1 and 3.5.2).

3.5.1 Marking on identification plate. Marking on identification plate shall consist of the manufacturer's type of model number, serial number, contract or order number, and date of manufacture. The salient features including size, weight, speeds, voltage, current and the national stock number.

3.5.2 Marking on information plate. Information plate shall be marked to include the following: "MACHINE NOT TO BE LOADED WITH MORE THAN RATED POUNDS (DRY WEIGHT) OF CLOTHING."

3.6 Technical data. The contractor shall prepare manuals in accordance with the data ordering documents (see 6.1.2).

3.7 Workmanship. In general, workmanship shall be equivalent to the best in the industry and shall conform in detail to the following:

(a) Welds are cleaned and ground to present a uniform and smooth finish.
(b) All parts are tight and of rigid construction so that the machine does not leak or vibrate during operation.
(c) Cylinder, with load and without, does not wobble during operation.
(d) Access plates and doors are provided as necessary for proper maintenance, cleaning and lubrication, and are of suitable size and properly located.
(e) Controls and door handle are suitably located and hazard free to operator.
(f) All surfaces are free of burrs and sharp edges.
(g) Safety guards are adequate and suitably located.
(h) Drive adjustment mechanism is suitable for the purpose intended.
(i) Bearings, gears and shafts are properly designed, alined and fitted for durable operation.
(j) Lubrication fittings are fitted where necessary and are adequate for purpose intended.
(k) Fastenings are adequate and accomplish the purpose for which intended.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

(a) First article inspection (see 4.3).
(b) Quality conformance inspection (see 4.4).

4.3 First article inspection. First article inspection shall consist of the examinations in table I and tests specified in 4.3.2 through 4.3.5.

4.3.1 Sample machine. When specified in 6.1.1 and prior to the start of production within the number of days specified in 6.1.1, one machine shall be made ready for approval of the contracting officer or his authorized representative for tests and examinations under 4.3.2 through 4.3.5.

4.3.2 Inclination test. The machine shall be inclined at an angle of 12 degrees to each side of two vertical planes at right angles to each other and shall operate with capacity load for a full cycle in each of the four positions. Water spillage shall not exceed one gallon during the test, and the spillage shall not come in contact with electrical parts (see 3.4.1).

4.3.3 Airborne and structureborne acoustical tests; type I washer-extractor only. The washer-extractor shall be tested in accordance with MIL-STD-740, and as specified herein.

4.3.3.1 Test fixture and set up procedures. For the purposes of these tests the washer-extractor shall be specially mounted on a test fixture to simulate shipboard installation. Design plans for fabrication of the fixture and installation procedures shall be as specified (see 4.3.3.5).

4.3.3.2 Test conditions. For both airborne and structureborne noise tests the machine shall be loaded with 10 pounds (dry weight) of test garments and noise monitored in all modes of machine cycle. The noise level plots shall include wash, rinse, extract and the fill modes with both inlet valves open. The entire test shall then be repeated with a 16 pound (dry weight) test load. The test garments shall contain 80 percent cotton minimum.

TABLE I. Classification of defects.

Examine	Defect (requirement paragraph)	Classification Major	Minor
Materials	Construction materials not as specified (see 3.2).	X	
Design, general	Design not as specified (see 3.3 and 3.3.1).	X	
Automatic programmed	Cannot provide control as specified (see 3.3.2.11).	X	
Water temperature control	Not as specified (see 3.3.2.12).	X	

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TABLE I. Classification of defects. - Continued

Examine	Defect (requirement paragraph)	Classification Major	Minor
Water lever controls	Does not provide for two levels (see 3.3.2.13).	X	
Door interlock safety devices	Not provided, not as specified (see 3.3.4.1).	X	
Emergency stop button/switch	Not provided, not as specified (see 3.3.4.2).	X	
Capacity, size, weight, water useage	Machine not designed for 16 pounds dry weight capacity (see 3.3).	X	
	Machine exceeds dimensions and weight limits (see 3.3.2.1 and 3.3.2.2).	X	
	Machine water consumption exceeds limits (see 3.3.2.3).	X	
Construction/ mechanical	Mounting holes not provided, not as specified (see 3.3.2.4).	X	
	Shell has openings for only valves and overflow vent (see 3.3.2.5.1).	X	
	Drain valve not as specified (see 3.3.2.8).	X	
	Piping, valves and fittings, not as specified (see 3.3.2.9 and 3.3.2.9.1).	X	
	Inlet valves are not 3 inches minimum above the maximum high water level (see 3.3.2.9.1).	X	
	Overflow connection not as specified (see 3.3.2.9.2).	X	
	Cylinder bearings and seals not readily replaceable (see 3.3.2.10.1).	X	
	V-belts not properly sized or not readily adjustable (see 3.3.2.10.2).	X	
	Grease line extension lines not provided (see 3.3.2.14).	X	

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TABLE I. Classification of defects. - Continued

Examine	Defect (requirement paragraph)	Classification Major	Classification Minor
Construction/ electrical	Motor not as specified (see 3.3.3.1).	X	
	Electrical connection boxes are not dripproof enclosed or better (see 3.3.3.2).	X	
	Electrical terminal blocks not marked to correspond with electrical diagram (see 3.3.3.3).	X	
	Wires not properly marked to correspond with electrical diagram (see 3.3.3.3).	X	
	Length of automatic control panel cable not as specified (see 3.3.1).	X	
Safety features	Belts, shafts, pulleys not properly enclosed (see 3.3.4).	X	
	Door interlock safety device not provided, or as specified (see 3.3.4.1).	X	
Identification	Identification and information plates not of specified material (see 3.5).		X
	Marking illegible (see 3.5).	X	
	Identification and information plates not provided, as specified (see 3.5.1 and 3.5.2).	X	
Workmanship	Surfaces of cylinder are not free of burrs and sharp edges (see 3.3.2.6).	X	
	Rust, toolmarks, pits, scratches, wrinkles are present (see 3.7).		X
	Welds not buffed (see 3.7).		X
	Surfaces are not free from burrs, sharp edges (see 3.7).		X

4.3.3.3 Airborne noise test. The washer-extractor mounted on the special test fixture shall be airborne noise tested to the aforementioned test conditions in accordance with MIL-STD-740 using the procedures for sound pressure level measurements. In addition to the octave band sound pressure levels, the overall A-weighted sound pressure level shall be obtained at each microphone position for each of the machines operating modes.

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4.3.3.3.1 Sound pressure acceptance levels. For measurements taken as detailed above the octave band sound pressure levels and overall dBA level specified in table II shall be used as the basis for acceptance for airborne noise of the washer-extractor.

TABLE II. Washer-extractor airborne noise acceptance levels (maximum).

Sound pressure, dB at 1 meter RE .0002 dyne/cm^2 for octave band center frequencies, Hz									Sound pressure, dBA at 1 meter RE 0.0002 dyne/cm^2
32	63	125	250	500	1K	2K	4K	8K	
73	69	71	63	62	62	59	58	59	66

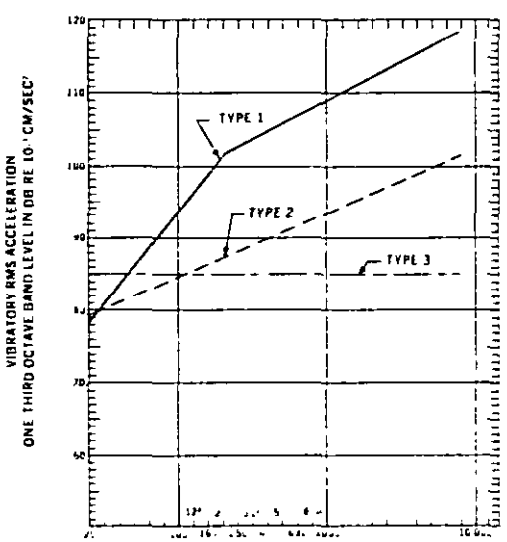
4.3.3.4 Structureborne noise test. The washer-extractor mounted on the test fixture shall be structureborne noise tested to the aforementioned test conditions in accordance with MIL-STD-740, except as specified herein.

4.3.3.4.1 Selection of transducer locations. Transducer triaxial blocks shall be attached to the supporting members of the machine at a minimum of two diagonally opposite corners as close as feasible to the machine mounting bolts. At least one of the corners selected shall be the one nearest the attachment for the drive motor.

4.3.3.4.2 One third octave band measurements. The noise level reported for each band shall be the maximum level appearing (excluding spurious signals) in any direction for each of the machine's acoustically significant operational modes (including extraction, deep rinse, and wash-and-fill) for a wash load of 10 pounds (dry weight) of clothes, and for 16 pounds full capacity-load.

4.3.3.4.3 Structureborne noise acceptance criteria. For measurements as taken above, the type 2 one-third octave band noise level acceptance line as specified in table III shall be used for acceptance criteria.

TABLE III. Band center frequency in hertz (one-third octave).



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4.3.3.5 Construction of test fixture and mounting boxes. Test fixture, equipment mounting boxes, and hold down straps shall be fabricated in accordance with figures 1 and 2.

4.3.3.6 Resilient mount pads. Resilient pads shall be chloroprene polymers (neoprene) in accordance with ASTM D 2000, composition 1BC615, 1/2 inch nominal thickness, 60 durometer hardness. Pads shall be fabricated as shown on figure 3.

4.3.3.7 Acoustical test set up procedure. The test fixture shall be firmly clamped to a structural concrete floor or machinery bedplate using six strap clamps. The mounting boxes and equipment shall be mounted as shown on figure 4 and affixed to test fixture as follows:

(a) Coat 3/8-16 jacking screw with release agent, and screw into equipment mounting box jacking-screw hole so that the tip is just shy of being flush with box bottom surface. The jacking screw is for removing the mounting boxes after the test.
(b) Clean the surface of the standard test fixture table and the bottom of the mounting boxes carefully of any grease or dirt using a suitable solvent.
(c) Clamp boxes into position on the damped table in the exact 'foot-print' pattern of the machine to be tested. Check the separation distances of the box holes to be sure their spacing matches exactly the spacing of the mounting holes in the machine in all directions: transverse, fore/aft, and diagonally.
(d) Clamp blocking-fences (or pairs of locating-stops) into position along two intersecting sides of the box.
(e) Remove box hold-down clamps and coat the bottom of box with epoxy, and comb to uniform distribution.
(f) Reclamp box in the proper position against the locating stops. Remove locating stops and jacking screws before epoxy fully cures.
(g) When epoxy fully cures, remove hold down clamps. Care must be taken not to subject the boxes to shock blows, nor allow heavy components to bump them, nor to subject them to undue tensile stresses by uneven bolting procedures, by bolting to machine feet that are substantially out of plane.
(h) Machine feet should be level and in place within 0.015 inch before inserting rubber pads. Correct with annular shims.
(i) If feet are appreciably tilted or out of plane, standard 7/8- washers may be epoxied to mounting points to achieve acceptable alinement.
(j) Hold down nuts shall be tightened evenly and gradually to compress the resilient pads to a final installed height that is 73-75 percent of their initial free height.

4.3.4 Extract efficiency test. Water retention of material shall not exceed 80 percent after 5 minutes of extraction. The test material shall contain 80 percent cotton minimum, and a minimum of 10 runs shall be performed to evaluate the performance of the machine under this requirement.

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4.3.5 High impact shock test. The washer-extractor and automatic control programmer (see 3.4.4) shall be tested in accordance with MIL-S-901 and as specified herein:

(a)	Shockproofness	- shall be grade B.
(b)	Required classes	- shall be class 1.
(c)	Test classifications	- shall be lightweight and mediumweight for the automatic control programmer, and washer-extractor, respectively.
(d)	Test type	- shall be type A.
(e)	Failure definition	- any part of test items coming adrift or separating from test specimen shall constitute failure of test.
(f)	Mounting	- washer-extractor shall be tested on fixture shown on figure 10-1 of MIL-S-901. Automatic control programmer shall be tested on fixture 4-A as shown on figure 5 of MIL-S-901.
(g)	Mode	- equipment not operating or energized.
(h)	Disposition	- ordering data (see 6.1.1).
(i)	Quantity for test	- 1 each washer-extractor, and 1 each automatic control programmer.

4.4 Quality conformance inspection.

4.4.1 Inspection lot. A lot shall consist of all machines of one design offered for delivery at one time.

4.4.2 Sampling for visual and mechanical inspections. Examination of the end item shall be for defects specified in table I. The inspection level and Acceptable Quality Level (AQL) shall be in accordance with MIL-STD-105, level II with an AQL of 1.5 for major defects and 6.5 for minor defects.

4.4.3 Lot tests. Each sample selected in accordance with 4.4.2 shall be subjected to the following tests.

4.4.3.1 Operational testing. Each washer-extractor furnished under the specification shall be connected to electrical power supply and water, and operationally checked as follows:

(a) Automatic controller properly and automatically controls all sequences of operation in proper order.
(b) Door safety lock prevents opening of door during any phase of operation.
(c) No leakage is present in piping, shell, and door components.
(d) Cylinder reverses minimum three times per minute.
(e) Indicating lights, switches, pushbuttons and timers function properly.

Failure to pass any of the above shall be cause for rejection of the unit.

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4.4.3.2 Insulation testing. Insulation resistance test shall be conducted on each machine. Insulation resistance of motor, controller, and connecting cable shall be measured by a 500 volt direct current insulation resistance tester. Insulation resistance for each component shall not be less than 10 megohms.

4.4.4 Test reports and certifications. The contractor shall certify in writing that the materials used in the washer-extractors are as specified herein. The contractor shall also certify that the visual, operational, and insulation resistance tests and examinations have been performed. When testing is performed, the contractor shall document the test results and submit test reports in accordance with data ordering documents (see 6.1.2). Test conditions shall include the following conditions under which equipment was tested:

(a) Water line pressures.
(b) Composition of test garments.
(c) Type of inlet valve construction.

4.5 Inspection of packaging. Sample packages and packs and the inspection of the packaging, packing, and marking for shipment and storage shall be in accordance with the requirements of section 5 and the documents therein.

5. PACKAGING

(The packaging requirements specified herein apply only for direct Government acquisitions. For the extent of applicability of the requirements of referenced documents listed in section 2, see 6.3.)

5.1 Packaging (levels A, B or C). Washer-extractors shall be preserved and packaged, level A or C, packed level A, B or C as specified (see 6.1.1) and marked for each level of protection in accordance with the applicable requirements of MIL-L-3153 (see 6.1.1). Washer-extractors shall be shipped fully assembled, unless disassembly is necessary to protect certain components during transit, or disassembly results in lower transportation cost to the Government or to the contractor.

6. NOTES

6.1 Ordering data.

6.1.1 Acquisition requirements. Acquisition documents should specify the following:

(a) Title, number and date of this specification.
(b) Specify whether machine is for type I or type II use (see 1.2).
(c) If first article is required (see 3.1).
(d) Number of days allowed for preparation for first article sample (see 4.3.1).
(e) Electrical characteristics (see 3.3.3.1).
(f) Preservation, packaging, packing, and marking requirements other than those required (see 5.1).
(g) Disposition of first article test equipment (see 4.3.5).

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6.1.2 Data requirements. When this specification is used in an acquisition which incorporates a DD Form 1423, Contract Data Requirements List (CDRL), the data requirements identified below shall be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved CDRL incorporated into the contract. When the provisions of DAR 7-104.9 (n)(2) are invoked and the DD Form 1423 is not used, the data specified below shall be delivered by the contractor in accordance with the contract or purchase order requirements. Deliverable data required by this specification is cited in the following paragraphs.

Paragraph no.	Data requirement title	Applicable DID no.	Option
3.6	Manual, technical preliminary	DI-M-2043	10.2
3.6	Commercial manual	DI-M-5104	10.1
4.4.4	Report, component shop noise tests	UDI-T-23764	----
4.4.4	Certification of compliance	DI-E-2121	10.1
4.4.4	Procedures, test	UDI-T-23732	10.2
4.4.4	Report, test/inspection	UDI-T-23473	----

(Data item descriptions related to this specification, and identified in section 6 will be approved and listed as such in DoD 5000.19L., Vol. II, AMSDL. Copies of data item descriptions required by the contractors in connection with specific acquisition functions should be obtained from the Naval Publications and Forms Center or as directed by the contracting officer.)

6.1.3 The data requirements of 6.1.2 and any task in section 3, 4, or 5 of the specification required to be performed to meet a data requirement may be waived by the contracting/acquisition activity upon certification by the offeror and accepted by the Government under a previous contract for identical item acquired to this specification. This does not apply to specific data which may be required for each contract regardless of whether an identical item has been supplied previously (for example, test reports).

6.2 First article inspection. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection as to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract.

6.3 Sub-contracted material and parts. The preparation for delivery requirements of referenced documents listed in section 2 do not apply when material and parts are acquired by the contractor for incorporation into the equipment and lose their separate identity when the equipment is shipped.

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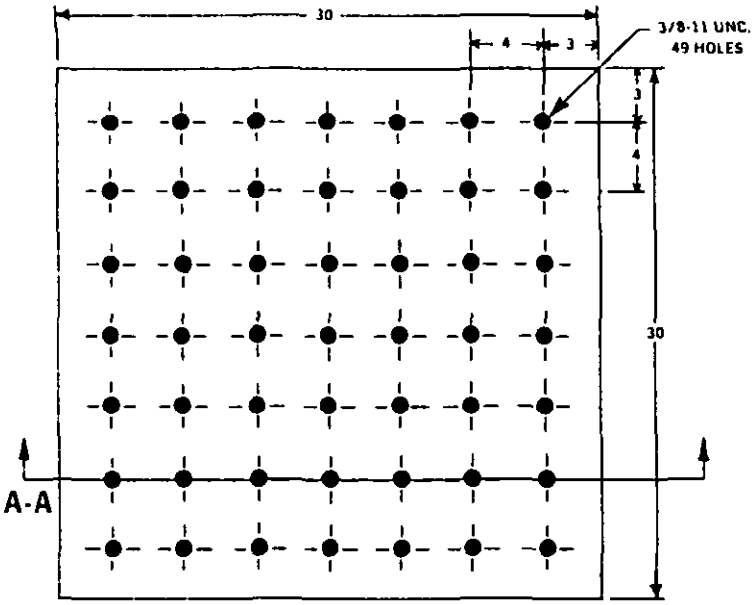
6.4 Provisioning. Provisioning Technical Documentation (PTD), spare parts, and repair parts should be furnished as specified in the contract.

6.4.1 When ordering spare parts or repair parts for the equipment covered by this specification, the contract should state that such spare parts and repair parts should meet the same requirements and quality assurance provisions as the parts used in the manufacture of the equipment. Packaging for such parts should also be specified.

6.5 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

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Navy - SH
(Project 3510-N290)

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THE 49 TAPPED HOLES ARE OPTIONAL, ONLY THOSE NECESSARY FOR SECURING THE MOUNTING BOXES HOLD DOWN CLAMPS ARE REQUIRED. DIMENSIONAL LOCATION OF HOLES IS UNCHANGED.

NOTES:

1. MATERIAL: STEEL
2. TOLERANCES: ±1/16 EXCEPT WHERE NOTED
3. AFTER MACHINES, INSERT BOLTS INTO NINE TAPPED HOLES CENTERED OVER CAVITIES. FILL CAVITIES WITH "SACRETE BLACKTOP" OR EQUIVALENT AND PACK SOLIDLY.

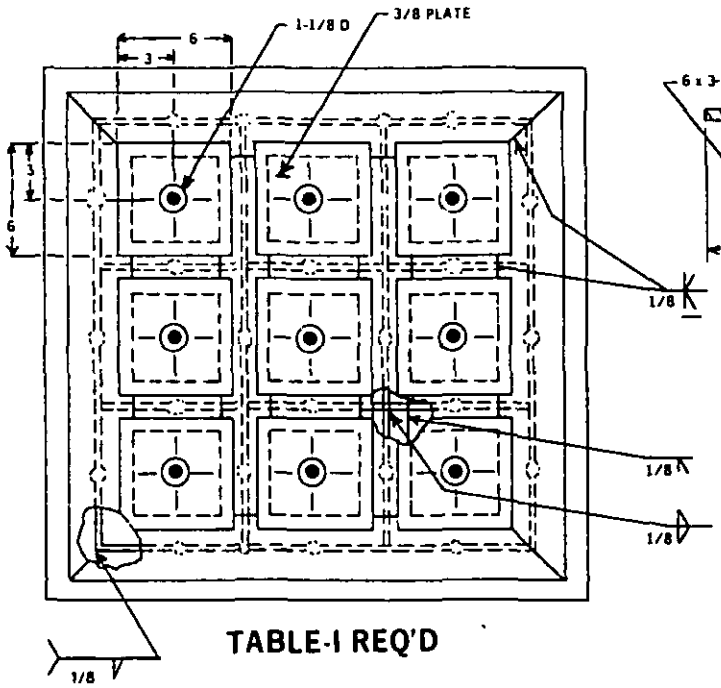
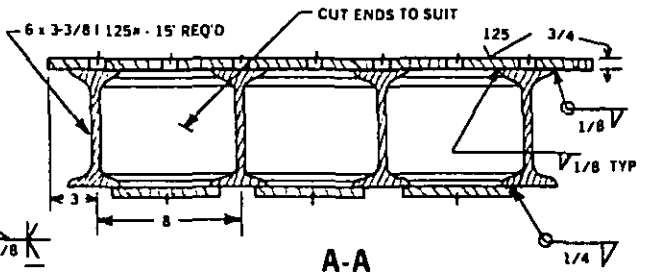
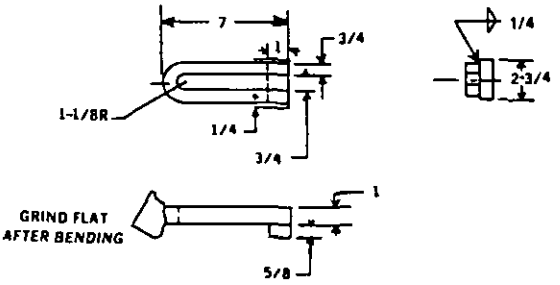


TABLE-I REQ'D



A-A



CLAMP-8 REQ'D

SH 12420

FIGURE 1. Standard test fixture and clamps.

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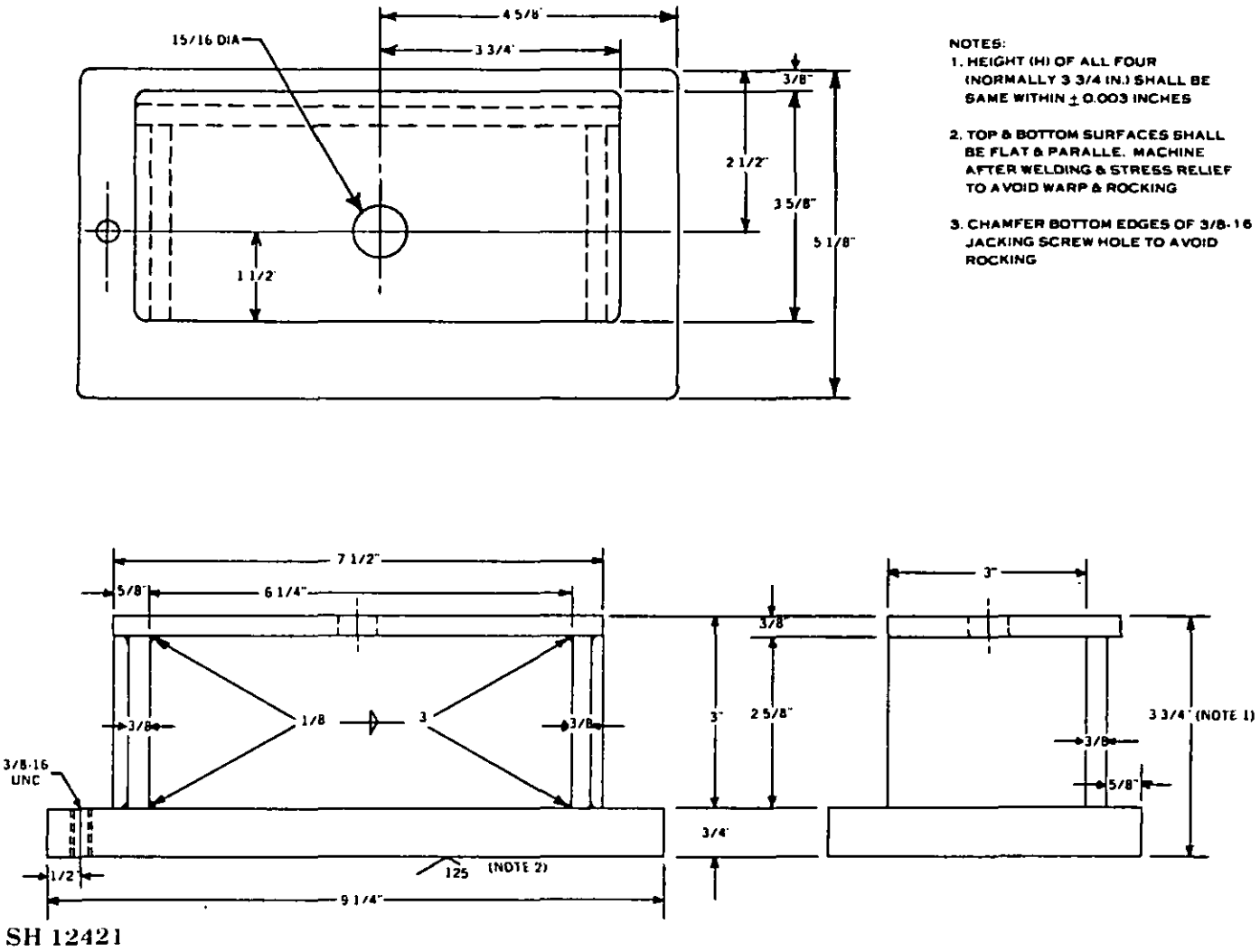


FIGURE 2. Mounting boxes - steel, 4 - required.

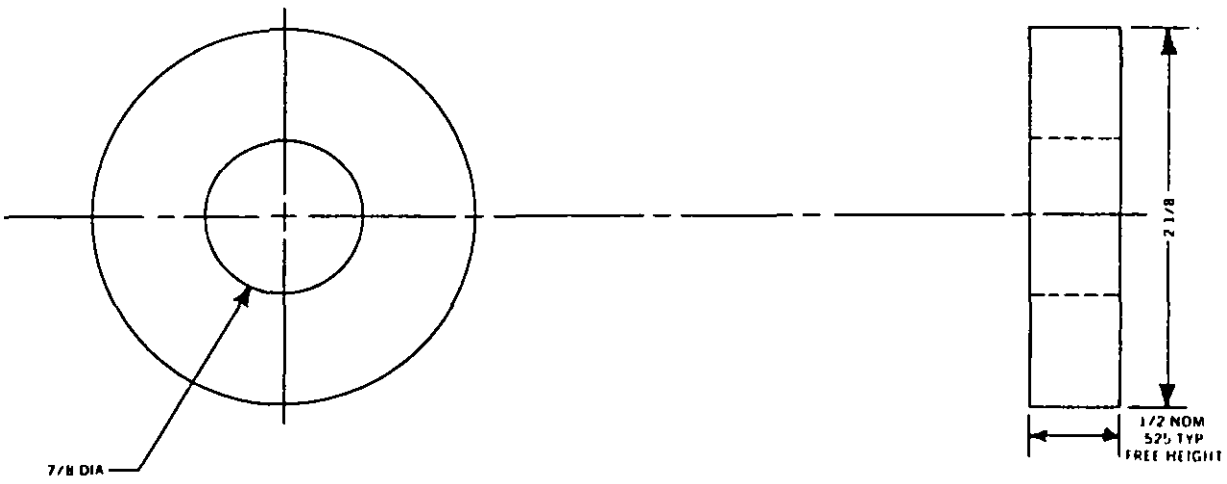
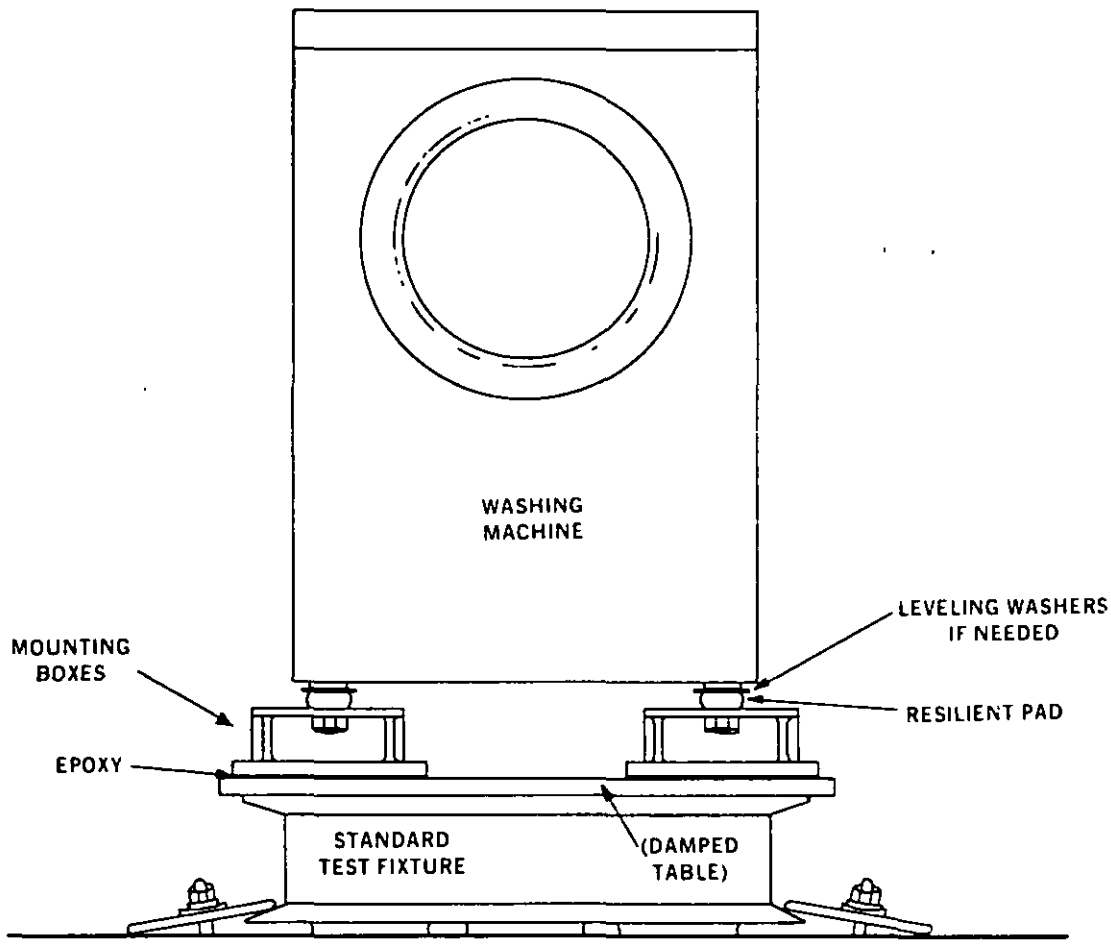


FIGURE 3. Resilient pads, 4 - required.

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SH 12423

FIGURE 4. Test fixture and item assembly.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-W-23554C(SH)	WASHER-EXTRACTOR, LAUNDRY, COMMERCIAL (SHIPBOARD)

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*: _____
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

DD FORM 82 MAR 1426 PREVIOUS EDITION IS OBSOLETE.

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